

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Regular/Supplementary Winter Examination – 2024**

Course: B.Tech. **Branch :** Electronics Engineering/ Electronics & Telecommunication Engineering /
Electronics and Communication Engineering **Semester : VII**

Subject Code & Name: Wireless Sensor Networks BTEXOE703A/BTETOE703A

Max Marks: 60

Date:10/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks				
Q. 1	Objective type questions. (Compulsory Question)		12				
1	What is wireless communication?		1				
	<table><tr><td>a. Sending data from one location to with the use of physical medium</td><td>b. Sending data from one location to another without the use of physical medium</td><td>c. Sending data from one location to another without the use of virtual medium</td><td>d. None of the mentioned</td></tr></table>	a. Sending data from one location to with the use of physical medium	b. Sending data from one location to another without the use of physical medium	c. Sending data from one location to another without the use of virtual medium	d. None of the mentioned		
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2	Which of the following is a type of wireless communication?		1				
	<table><tr><td>a. LAN</td><td>b. WAN</td><td>c. PAN</td><td>d. All of the mentioned</td></tr></table>	a. LAN	b. WAN	c. PAN	d. All of the mentioned		
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3	Which of the following specifies a set of media access control (MAC) and physical layer specifications for implementing WLANs?		1				
	<table><tr><td>a. IEEE 802.11</td><td>b. IEEE 802.16</td><td>c. IEEE 802.15</td><td>d. IEEE 802.3</td></tr></table>	a. IEEE 802.11	b. IEEE 802.16	c. IEEE 802.15	d. IEEE 802.3		
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4	A wireless network without a centralized access point may be		1				
	<table><tr><td>a. Infrastructure network</td><td>b. Ad-hoc network</td><td>c. All of these</td><td>d. None of these</td></tr></table>	a. Infrastructure network	b. Ad-hoc network	c. All of these	d. None of these		
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5	Which of the following technologies enable wireless ad-hoc networking.		1				
	<table><tr><td>a. Zigbee</td><td>b. 4G LTE</td><td>c. All of these</td><td>d. None of these</td></tr></table>	a. Zigbee	b. 4G LTE	c. All of these	d. None of these		
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6	The main source of power consumption in wsn is due to		1				
	<table><tr><td>a. Sensing</td><td>b. Transmitting</td><td>c. Processing</td><td>d. None of these</td></tr></table>	a. Sensing	b. Transmitting	c. Processing	d. None of these		
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7	RTS/CTS period is called					1
	a.Waiting period	b. Contention period	c. Running period	d. None of these		
8	Existing MAC protocol can not be used in MANETs because					1
	a. Resource constrained nodes	b. Limited Bandwidth	c. Lack of centralized control	d. All of the above		
9	Nodes which deliberately disrupt the correct operation of the routing protocol are known as					1
	a.Selfish nodes	b. Failed nodes	c. Badly failed nodes	d. Malicious nodes		
10	In MACAW data sending frame is of size					1
	a.120 Bytes	b. 30 Bytes	c. 23 Bytes	d. 35 Bytes		
11	Which is hybrid protocol					1
	a. DSR	b. AODV	c. ZRP	d.None of these		
12	When a group of nodes are required to contact, which type of the following routing is useful.					1
	a. Multicast	b. Unicast	c. Broadcast	d. None of these		
Q. 2	Solve the following.					12
A)	Define the term Wireless sensor network? Draw and explain the architecture of WSN?					6
B)	What are the various constraints and challenges for WSN.					6
Q.3	Solve the following.					12
A)	Draw and explain the MANET node model.					6
B)	What are the various factors that affect the MANET.					6
Q. 4	Solve Any Two of the following.					12
A)	Describe MAC protocol in brief.					6
B)	What is Routing in WSN? Explain the routing process in WSN?					6

C)	Describe the STEM (Sparse topology and energy management) in brief.	6
Q.5	Solve Any Two of the following.	12
A)	What is Gateway? What is the need of Gateway in WSN?	6
B)	Explain the various components of sensor node and give their functionality.	6
C)	Explain the different sources of power available for WSN?	6
Q. 6	Solve Any Two of the following.	12
A)	Write short notes on: 1.TinyOS 2. MANTIS	6
B)	Describe the functionality of Sensors and Actuators in brief.	6
C)	Write short notes on: 1. LEACH 2. ZIGBEE	6
	*** End ***	